

ITU-MiniTwit — BSc DevOps, Software Evolution and Software Maintenance

Group: BSc_group_m

Repository: <https://github.com/RonoITU/itu-devops2026-jackhammers>

Issue tracker: <https://github.com/RonoITU/itu-devops2026-jackhammers/issues>

Monitoring dashboard: <http://178.104.27.224:3000/d/chirp-aspnet-001/windysquirrels-monitoring-d>

Logging dashboard: <http://178.104.27.224:3000/d/app-logging-dashboard/windysquirrels-logging-d>

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1. System's Perspective

1.1 Design and Architecture

Author(s): Rasmus Alexander Christiansen

This section describes the overall system architecture of MiniTwit, including how the application is structured, deployed, and monitored. The diagram above provides a visual overview of the components and their interactions across the infrastructure.

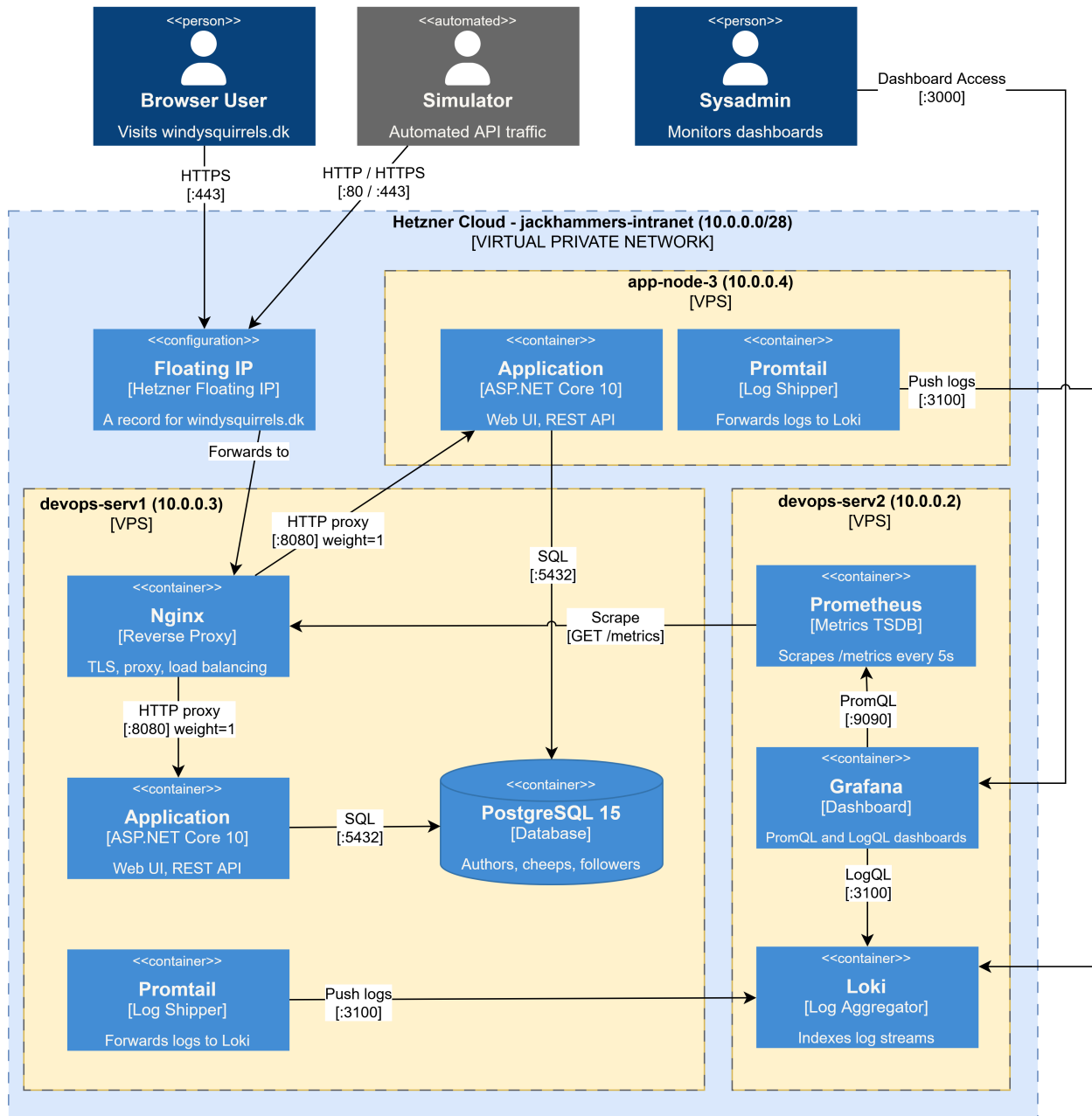


Figure 1: Diagram of the system architecture

MiniTwit is deployed across three Hetzner Cloud VPS nodes connected via a virtual private network

(10.0.0.0/28).

devops-serv1 (10.0.0.3) is the primary node. Nginx runs here as the sole public entry point, handling TLS termination via Let's Encrypt, redirects HTTP traffic to HTTPS, and weighted load balancing across both app nodes. The primary application instance serves web UI and REST API traffic and exposes a /metrics endpoint for Prometheus. PostgreSQL 15 hosts the shared database for all application data. Promtail runs as a log shipping agent, collecting Docker container logs from this node and forwarding them to Loki on devops-serv2.

app-node-3 (10.0.0.4) is the secondary application node. It runs an application replica that receives the majority of traffic from Nginx (weight=2) and connects to the shared PostgreSQL database on devops-serv1. Promtail runs here as well, shipping container logs to Loki on devops-serv2.

devops-serv2 (10.0.0.2) is dedicated to observability. Prometheus scrapes the /metrics endpoint on devops-serv1 every five seconds and stores the resulting time-series data. Loki aggregates the structured log streams pushed by the Promtail agents on devops-serv1 and app-node-3. Grafana provides dashboards over both data sources using PromQL for metrics and LogQL for logs.

1.2 Dependencies

Author(s): Rasmus Alexander Christiansen

The diagram below illustrates the relationships between the key technologies across all layers of the system. Not every dependency is shown - minor libraries and transitive dependencies are omitted for clarity. A full list with descriptions is provided in the tables below the diagram.

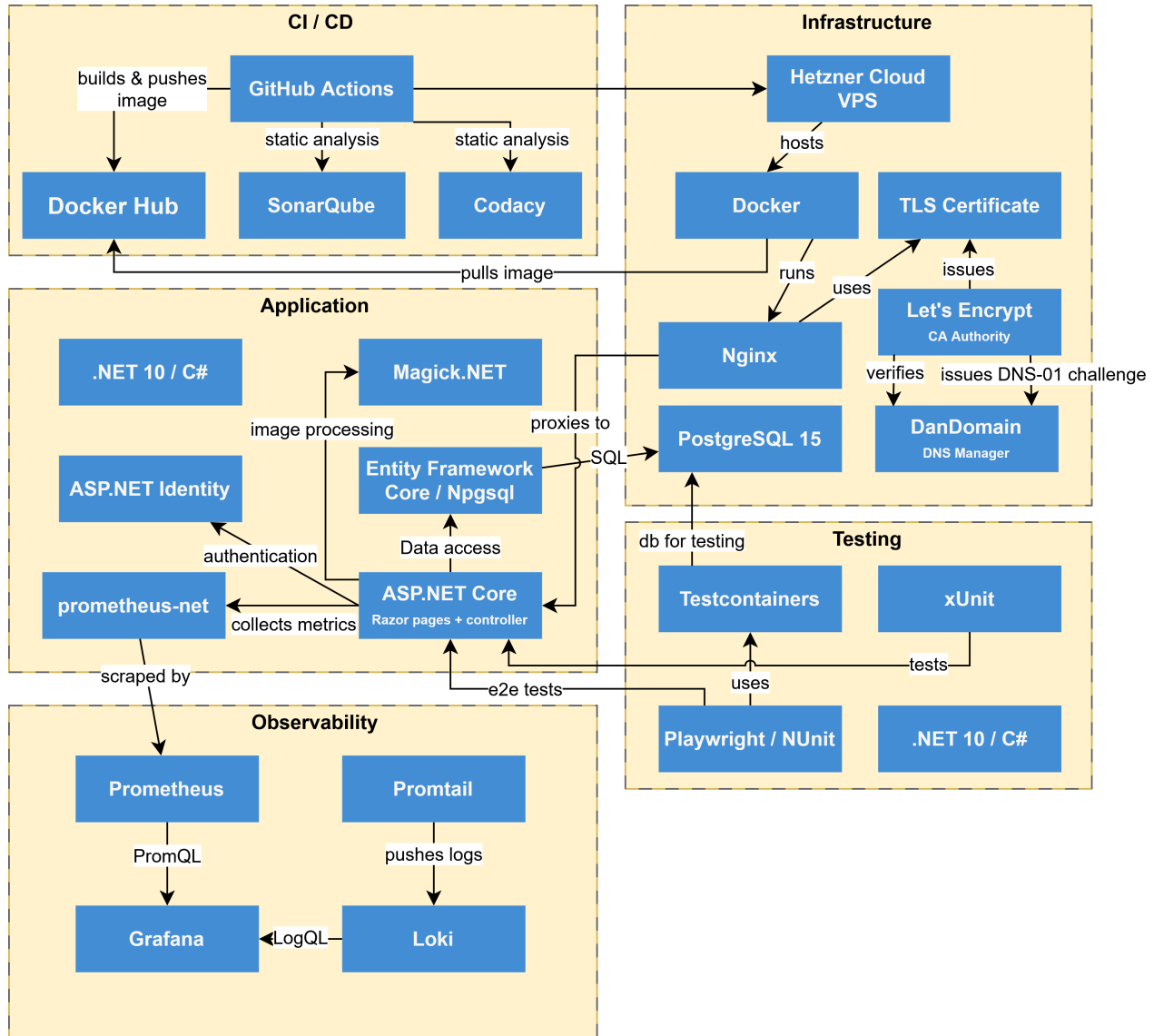


Figure 2: Diagram of the system dependencies

Application

Technology / Tool	Purpose
C# / .NET	Application language and runtime
ASP.NET Core (Razor Pages)	Web UI framework
ASP.NET Core (Controllers)	REST API for the simulator
Entity Framework Core	ORM and database migrations
Npgsql EF Core Provider	PostgreSQL driver for EF Core
ASP.NET Core Identity	User authentication and account management
prometheus-net	Exposes application metrics to Prometheus
Magick.NET	Profile image processing
SixLabors.ImageSharp	Image resizing and encoding

Infrastructure

Technology / Tool	Purpose
Hetzner Cloud	VPS hosting for all three nodes
Docker	Container runtime on all nodes
Docker Compose	Service orchestration on each node
Nginx	Reverse proxy, TLS termination, load balancing
PostgreSQL	Relational database
Let's Encrypt / Certbot	Automated TLS certificate provisioning

Observability

Technology / Tool	Purpose
Prometheus	Metrics scraping and time-series storage
Grafana	Dashboards for metrics and logs
Loki	Log aggregation and indexing
Promtail	Log shipping agent (Docker container logs)

CI/CD

Technology / Tool	Purpose
GitHub Actions	CI/CD pipeline automation
Docker Hub	Container image registry
SonarQube (SonarCloud)	Static analysis and quality gate
Codacy	Additional static analysis
Pandoc + XeLaTeX	Report PDF generation

Testing

Technology / Tool	Purpose
xUnit	Unit and integration test framework
NUnit + Playwright	End-to-end browser testing
Testcontainers	Spins up a PostgreSQL instance for tests
coverlet	Code coverage collection

1.3 Current State of the System

Author(s): Rasmus Alexander Christiansen

The current state of the system is assessed through two static analysis tools integrated into the CI pipeline: SonarCloud and Codacy. Both run automatically on every pull request. Rather than blocking merges on quality gate failure (**continue-on-error: true** in the pipeline), the team has deliberately prioritised visibility over enforcement - the goal being that issues are made transparent and trackable, not that velocity is sacrificed to a strict gate.

SonarCloud

SonarCloud provides a continuous quality assessment of the codebase across security, reliability, maintainability, coverage, and duplications.



Figure 3: SonarCloud project summary

It is worth noting that the project was inherited from a 1.5 year old codebase originally written by students in the third-semester BDSA course. As a result, a significant portion of the issues identified by SonarCloud reflect legacy code rather than code introduced during this course. The team has primarily focused on resolving security issues and issues affecting performance, while reliability, maintainability, and code duplication issues in the inherited code have not been a priority.

The overall results show 1 security issue, 41 reliability issues, 106 maintainability issues, 75.4% test coverage, and 8.7% code duplication. There are 0 security hotspots and all hotspots have been reviewed.

The Issues view illustrates what this visibility looks like in practice:

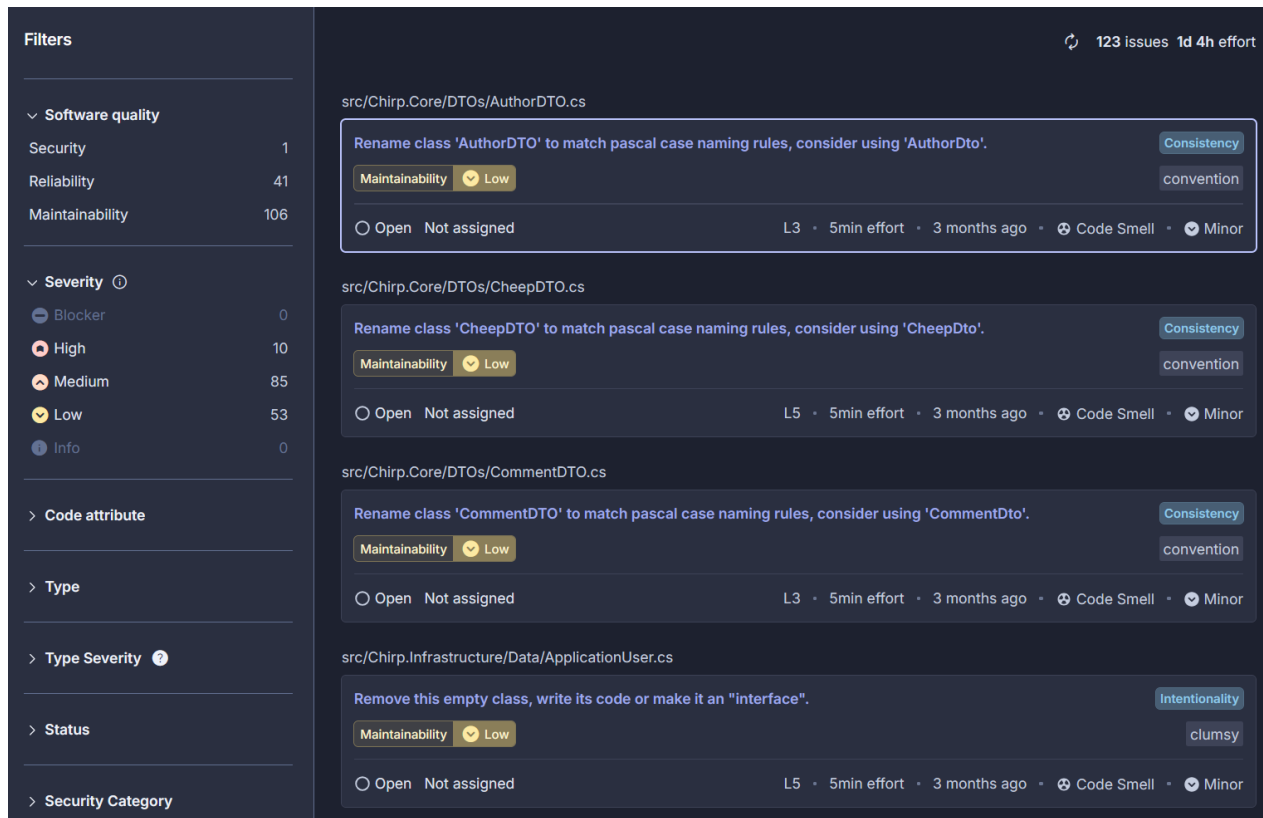


Figure 4: SonarCloud issues

SonarCloud surfaces concrete, prioritised issues directly tied to source locations - for example, flagging that `AuthorDTO` in `src/Chirp.Core/DTOs/AuthorDTO.cs` should be renamed to `AuthorDto` to conform to Pascal case naming conventions. Before static analysis was introduced, issues like this existed in the codebase but were invisible. They are now explicit, categorised by severity, and can be worked through systematically.

Codacy

Codacy runs as a secondary static analysis tool in the CI pipeline alongside SonarCloud, providing an independent second opinion on code quality. It does not currently expose a public dashboard, but its results are available in the pipeline logs on every run.

2. Process' Perspective

2.1 CI/CD Pipeline

*Author(s): Ronas Olsen and Jacob Sponholtz

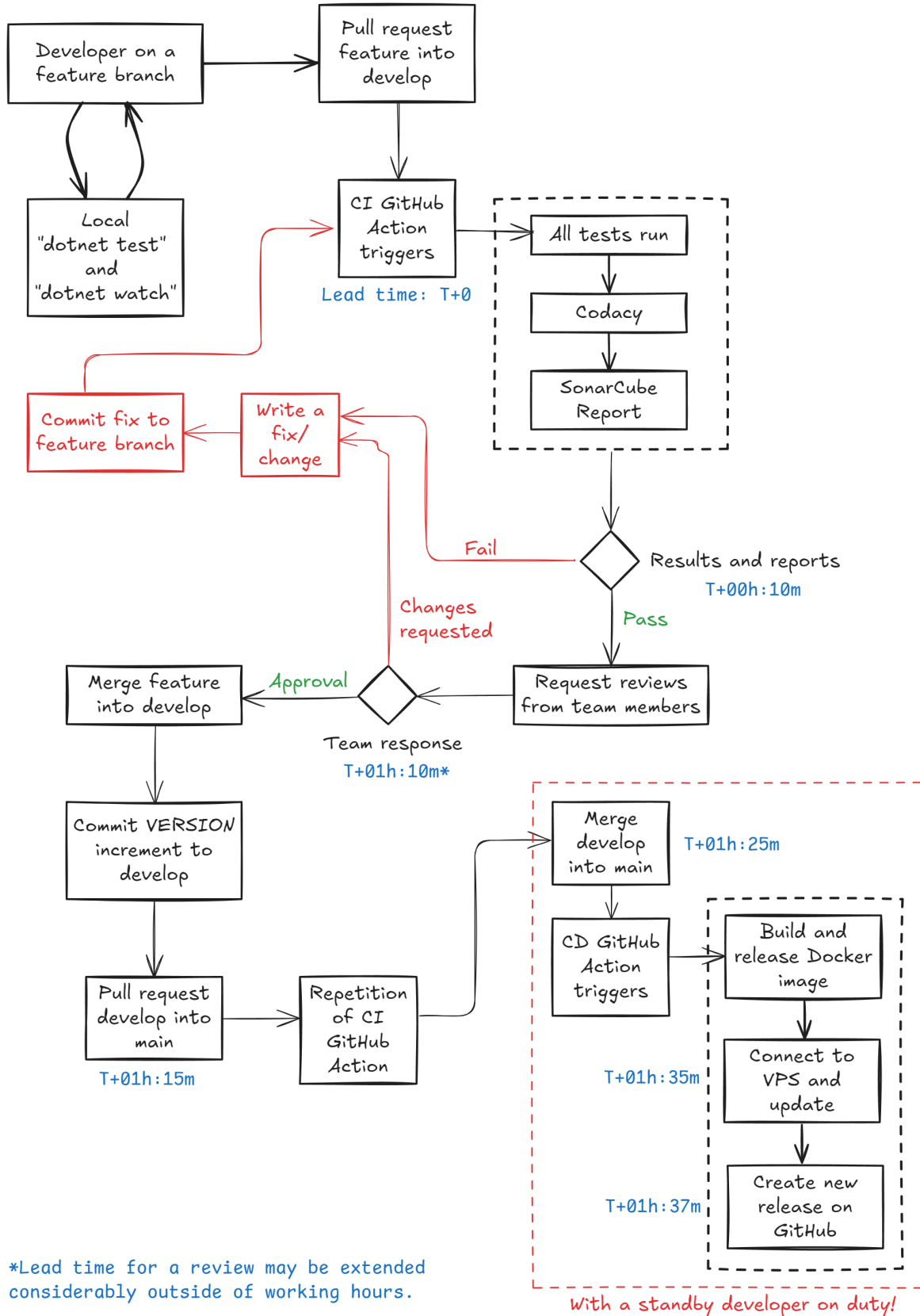


Figure 5: Diagram of the CI/CD Pipeline (excalidraw.com)

The source code is hosted on GitHub. We follow a modified GitFlow branching strategy: Feature development on a `develop` branch and feature branches, but without the use of release branches. Instead `develop` is merged directly to the main branch on release, and all QA is handled as part of the CI of features to the `develop` branch.

The main branch is the version currently in production. Versions are also tagged semi-automatically by the CD workflow with each new release.

Any contributor can clone the repository, create a new feature branch, and work using their preferred tools. Simple instructions are provided in the README on how to build, run and test locally. To integrate a feature into the next release, the contributor opens a pull request towards `develop`. This triggers the CI GitHub Action to run our testing suite (Unit, Integration, E2E) and static analysis tools (SonarCube, Codacy) so that we will have concrete QA evidence alongside the changes to review.

The contributor will also get immediate feedback from the analysis tool, indicating any new issues and the test coverage on new code. Once the contributor is happy with the new feature(s), we ask for the feature branch to be merged.

To deploy a new release, the developer must first assert the new commits on `develop`, increment the VERSION file accordingly, and open a pull-request towards main. This pull-request first triggers the CI GitHub Action. When this action has been completed successfully, the developer can merge towards main. The merge triggers the CD actions, in which a new Docker image is built and uploaded to Docker Hub. Github then connects to the server via ssh to recreate and restart the containers, using the new release image.

Finally the CD action will tag the release with version number, and create a new release on GitHub.

2.2 Monitoring

*Author(s): Ronas Olsen, Jacob Sponholtz and Rasmus Alexander Christiansen

System monitoring is based on Prometheus and Grafana. Prometheus collects metrics from app containers over the VPN. Grafana is setup to poll for this data as needed for dashboard information on our app service. This is done on a system monitoring dashboard and a business intelligence (BI) dashboard.

The most important metrics on the monitoring dashboard are HTTP traffic, error rate, request rate, latency, memory and CPU usage. In the BI dashboard, the highlights are a top 10 most followed, total users versus active users, total messages and average followers per user. (See [logging-dashboard.gif](#) and [monitoring-dashboard.gif](#) for reference.)

2.3 Logging

*Author(s): Ronas Olsen, Jacob Sponholtz and Rasmus Alexander Christiansen

Logs are collected from Docker containers on each VPS using Loki and Prometheus. The standard outputs from each container are directed to an instance of Prometheus. Loki running on the monitoring server collects data from these for Grafana to access as needed for dashboards or for Drill-down. All log entries have a UTC timestamp appended automatically. (See [logging-dashboard.gif](#) for reference.)

2.4 Security Hardening

Author(s):

2.5 Availability and Scaling

Author(s):

3. Reflection Perspective

3.1 Evolution and Refactoring

Author(s): Jakob Sønder

Migrating away from Python Flask: This was solved in PR [#2](#) and [#3](#). A working ASP.NET application meeting similar requirements was utilized ¹. All dependencies were upgraded to target the newest long-term ASP.NET version, .NET 10.

Switching from SQLite to PostgreSQL: This was solved in PR [#8](#). We wanted to prepare for the possibility of a production setup, where multiple containers or applications can connect to the same database over a network. The biggest challenge was that earlier EF Core migrations were created for SQLite specifically. This meant migrations had to start over. Since the switch was done before production, major issues were avoided (see [docs](#) for solution considerations, if we had been in production).

3.2 Operation

Author(s):

3.3 Maintenance

Author(s): Jakob Sønder

The biggest maintenance challenges encountered concerned surrounding systems intended to improve MiniTwit stability. An example was ensuring that coverage reports, integration tests and SonarCube agreed on where test results were generated and how they were reported (PR [#93](#), [#94](#), [#95](#)).

Another example, was maintenance challenges involving the move to a smaller Alpine-based runtime image for deployment (PR [#46](#)). This improved deployment size, but introduced maintenance issues around certificates, root access and timezone data (PR [#48](#)).

3.4 DevOps Reflection

Author(s):

¹link to ASP.NET project: <https://github.com/ITU-BDSA2024-GROUP7/Chirp>

4. Use of Generative AI

Author(s): Rasmus

GitHub Copilot was used to assist with boilerplate code generation throughout development. It provided inline auto-completion that often matched the intended behavior being implemented, and helped untangle unfamiliar functionality across the broader codebase. This sped up the coding process significantly, particularly for repetitive or structurally predictable code.

Microsoft Copilot was used for general consultation when extending the infrastructure, serving as a quick reference for exploring options and approaches before committing to a direction.

Claude was used for more in-depth technical discussions, typically involving specific code examples. It helped provide clarity on complex topics and was particularly useful when a concept required more thorough explanation than a quick search could offer.